Review of Deep Learning Models Based on Hyperspectral Image Classification Techniques in Monitoring Crop Production

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***Abstract*— the importance of evaluating and ensuring the quality of agricultural and horticultural products has increased in recent years. Conventional techniques that require Field measurements, research, as well as statistics analysis are expensive, and time taking labor-intensive. As a novel technology, HSI has been very popular, especially because of its potential uses in the field of remote sensing, particularly in agriculture. But, there are a number of difficulties in identifying HSI data, including the complex non-linear relationship between spectral bands and spatial coordinates, limited availability of training data and the massive duplication of spectral bands. Interestingly, Deep Learning (DL) techniques have demonstrated remarkable efficacy in a number of HSI analytic applications, including agricultural ones. There is an urgent need for a thorough survey that will help academics understand the significant progress made and the fascinating potential avenues for future research in this area, given the growing interest in using DL approaches to leverage HSI data for agricultural applications. This study of the literature carefully gathers, examines, and talks about recent attempts to use DL approaches. Autoencoders (AE), Recurrent Neural Networks (RNN), Deep Belief Networks (DBN), Generative Adversarial Networks (GAN), Convolutional Neural Networks (CNN) (in 1D, 2D, and 3D configurations), Transfer Learning (TL), Semi-Supervised Learning (SSL), Few-Shot Learning (FSL), and Active Learning (AL) are just a few of the techniques that fall under this broad category. These methods are designed to tackle the particular difficulties presented by HSI analysis in agriculture.**

The performance demonstrated by these various approaches is assessed and discussed in this review. In light of this, the effectiveness of these methods has been thoroughly examined and debated in light of the findings of cutting-edge research on well-known land cover datasets repository on Github. The viability, adaptability, and affordability of using hyperspectral imagery (HSI) for crop classification in agricultural regions have been extensively established.

Keywords*— *CNN, RNN, GAN, HSI, Deep learning, Hyperspectral imaging, Sustainable Agriculture

I. INTRODUCTION

The need to evaluate and protect the quality of agricultural and horticultural products has grown during the last 20 years. It has been demonstrated that traditional approaches, such as statistical analysis, field measurements, and investigations, are expensive, time-consuming, and labor-intensive (Ravikanth et al., 2017). Non-destructive and eco-friendly technologies have become more popular in response to these constraints. Hyperspectral Imaging (HSI) has therefore become a viable remedy. HSI makes it possible to take several pictures at different wavelengths (Chang, 2003), which makes it easier to precisely track changes in farms over time and space (Ge et al., 2016). This feature facilitates quick and precise forecasts of crop growth and improves agricultural decision-making. By combining spectroscopy and conventional imaging methods, Chemical and spectroscopic imaging, another name for hyperspectral imaging (HSI), is a revolutionary development. The simultaneous collecting of spectral and spatial information from an image is made possible by this fusion.

Although HSI technology was first developed for distant sensing, its benefits in outperforming conventional machine vision have been acknowledged in a number of fields, such as optical sensing and agriculture. Furthermore, as imaging techniques have advanced, HSI has become increasingly important for HSI gives farmers vital equipment that significantly improves their ability to make decisions. Its broad range of applications includes forecasting crop yields, identifying illnesses, and tracking crop health, growth, and resource allocation (Zhang et al., 2016; Li et al., 2020a).A more productive and environmentally friendly agriculture sector will eventually result from the numerous opportunities that AI and HSI present to transform management of fresh product quality and other important aspects of agriculture (He et al., 2019). This paper's main contribution is to examine how HSI is used in the agriculture industry by evaluating the efficacy and efficiency of deep learning methods applied to HSI data for practical uses. Insights into novel opportunities for creating intelligent systems in the fields of horticulture, agriculture, and food are presented in this study. It advances the industry's usage of artificial intelligence.

The primary contributions of this study can be summarized as follows:

- Providing the current situation of HSI in agriculture today, emphasizing noteworthy developments and identifying important areas that require more study.
- Examining a range of publicly accessible Agricultural datasets from HSI, emphasizing their distinctive characteristics and the state-of-the-art results obtained on every dataset.
- Examining several DL methods and strategies for HSI and offering a thorough evaluation of their advantages and disadvantages.

The primary uses of HSI technology in the agricultural sector are shown and discussed. This section offers

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information on how HSI can raise agricultural quality, safety, and production.

II. STATE-OF-THE-ART

Significant advancements have been made in the quickly changing field of HSI in agriculture, particularly in the areas of crop health monitoring and early plant disease identification using cutting-edge imagery and machine learning technologies. To further improve the capabilities of HSI technology and meet the evolving demands of the agricultural industry, it is critical to recognize and fill the current research gaps. Table 1 summarizes the results of several studies that provide insight into the present status of HSI applications in agriculture, emphasizing developments and pointing out relevant research gaps. By examining the uses of DL introduction and HSI in agriculture approaches, this study advances the field. This thoroughly investigates the efficacy of DL techniques using well-known datasets on land cover and closely examines different Applications of HSI in agriculture. By performing this, we identify the key research gaps, such as the requirement to determine the best techniques for DL and look into their most recent applications in the agricultural sector.

Application	Description	DL	Research Gaps	Reference
✓	Discussing geometric processing, sensor technology, measurement methodologies, and spectral remote sensing techniques using UAVs in conjunction with radiometric calibration considers the continuous advancements, possibilities, and constraints in the field of remote sensing.	×	Standard operating procedures should be established and distributed to ensure the quality of data. To minimize undesirable consequences during data collection and processing, it is important to comprehend the entire sensing process. Disseminating information on sensor technologies, data processing, and data acquisition techniques is necessary.	(Aasen et al., 2018), 2018
✓	Discussion about the developments in hyperspectral technology for the identification of plant diseases, with an emphasis on HSI applications and analysis. Highlighting the main obstacles facing the profession and recommending topics for more study.	×	-difficulties differentiating between biotic and abiotic stressors and detecting various infections. Plant disease early detection. The development of hyperspectral technologies based on satellites. More accurate agricultural information development technologies are required.	(Zhang et al., 2020a), 2020
✓	This investigation focuses on four distinct crop species and explores the most recent advancements in the use of hyperspectral remote sensing as a plant disease early detection technique. It highlights the considerable promise HRS offers for plant disease early detection diseases while also shedding light on the serious flaws in this field of study and outlining possible fixes.	×	Waveband interpretation heavily relies on plant physiology and biochemistry. In order to detect plant diseases using HRS with reliable results, a generic experimental approach is required.	(Terentev et al.2022)2022
✓	Explore the most recent deep learning methods that have been created specifically to assess and analyze HSI data in the agricultural industry assessing these techniques' efficacy using well-known land cover datasets.	✓	The best DL approaches for agriculture have been identified. An analysis of the most recent DL method implementations in agriculture	Ours, 2023

III. HSI DATASETS FOR ANALYSIS IN AGRICULTURE

A new era of applications in agriculture and research has been brought about by the introduction of HSI cameras, which has resulted in the labeling and collecting of large databases for agricultural research. The datasets have greatly increased the potential of machine learning methods, especially deep learning (DL), which needs a substantial quantity of data for training and testing. With 220 bands and a high spectral resolution, it is appropriate for in-depth spectrum investigation. It is better suited for more comprehensive LULC (Land Use Land Cover) study as opposed to fine-grained research due to its more modest spatial resolution of 20 m. For studies in general agriculture and comprehending wide patterns of LULC, this dataset is especially helpful. The dataset from Salinas (Scenes, 2020), which was further gathered by NASA AVIRIS in 1998, is more extensive, and comprises 54,129 samples from 16 classes. With a 3.7 m spatial resolution, it is perfect for intricate classification of crops jobs, and the exceptionally 224 bands of high spectral resolution is remarkable for thorough review. The Salinas dataset will be especially helpful to researchers seeking precision crop studies and in-depth agricultural class analysis. One noteworthy feature of the dataset from Pavia University (Scenes, 2020), which was gathered: The ROSIS-03 sensor was introduced in 2001; however, its size (42,776 samples) and lack of classes (9). For studies on urban agriculture, its 115-band intermediate spectral resolution is sufficient. The dataset's exceptional 1.3 m spatial resolution makes it ideal for in-depth research in urban settings. Particularly in smaller, more defined locations, this dataset is most appropriate for comprehensive land cover investigations and urban agriculture. Although they differ in size and diversity, the Chikusei (Yokoya and Iwasaki, 2016) and Botswana (Scenes, 2020) datasets, as well as According to Zhong et al. (2018), the WHU-Hi series The WHU-Hi-LongKou, WHU-Hi-HanChuan, and WHU-Hi-HongHu, are useful for crop categorization tasks. High resolutions of the spectrum (up to 240 bands in certain instances) provided by these datasets enable in-depth spectral research. Some datasets in the series

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WHU-Hi exhibit unusually high spatial resolutions (0.043–0.463 m range), although the resolution varies throughout these datasets. Although the Chikusei and Botswana databases are better suitable for classifying land cover and urban/rural region segmentation, specifically the WHU-Hi series is great for the classification of hyperspectral images and high-resolution crops. In fact, the wide variety of hyperspectral imaging datasets covered in this article serve as a basis in order to develop deep learning models. A wide variety of spectral data is offered by these datasets. This abundance of data helps models learn to recognize intricate patterns and minute details, improving their capacity for accurate analysis and responsible decision-making. Additionally, these datasets provide a thorough testing environment that makes it possible to assess and improve model performance in a variety of difficult situations. Essentially, these datasets support the creation of complex algorithms for deep learning and guarantee their dependability and applicability in actual agriculture settings.

IV. Approaches of Using Deep Learning to Analyze HSI

Over the past decade, deep learning techniques have dominated most computer vision research tasks, including hyperspectral imaging, because of their intrinsic mechanical capabilities for information representation that reflect steps like feature extraction (Sun et al., 2021a), selection (Zhao et al., 2019), and reduction of spatial-spectral and contextual features (Xu et al., 2019). Several reasons have contributed to this upswing: Innovative research in the nexus of computer science, agronomy, and remote sensing has been sparked by the growing interdisciplinary cooperation between these disciplines. The building of open-source deep learning libraries and potent computational resources such as PyTorch (Paszke et al., 2017), Keras [?], and TensorFlow (Abadi et al., 2015), along with the spread of sophisticated DL approaches and improved access to HSI data, have greatly reduced the entrance hurdles for researchers who want to use sophisticated models with little effort or money. The peak in 2023 may be attributed to advancements in the operational efficiency of hyperspectral sensors as well as a decrease in their physical footprint and cost. These advancements have expanded the usefulness Utilizing hyperspectral sensors for use in agriculture research and practical applications by allowing their integration into portable field instruments and drones, allowing for real-time in-situ analysis. Locating experts and promoting innovative research at the nexus of many disciplines. Furthermore robust machine learning, the availability of hyperspectral data, and open deep learning frameworks such as PyTorch (Paszke et al., 2017), Keras (Paszke et al., 2017), and TensorFlow (Abadi et al., 2015), as well as the advent of new deep learning techniques, which facilitated the implementation of intricate algorithms by academics without requiring a substantial amount of time or resources for development. The peak in 2022 will be reached when hyperspectral sensors become smaller and less expensive while also becoming more efficient. Their use in research and real-world agricultural settings has increased as a result of this development, which has made it easier to incorporate them into real-time monitoring tools for the field and drones.

A. CNN (Convolutional Neural Network)

CNNs are a revolutionary development in machine learning, particularly when it comes to processing and interpreting visual data (Bougourzi et al., 2020).

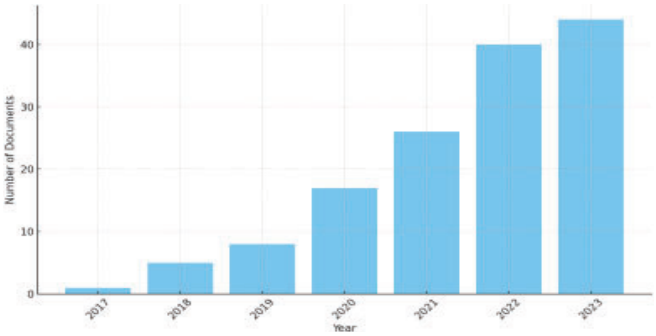

Fig. 1. The number of HSI articles published

CNNs mimic human visual information processing by taking influence from how the human visual cortex functions. They are crucial to HSI and particularly good at problems involving object identification (Ren et al., 2015), picture classification (Blaschke et al., 2014), and segmentation (Bougourzi et al., 2023a, 2023b). The special architecture of CNNs is made to effectively handle and analyze visual data.

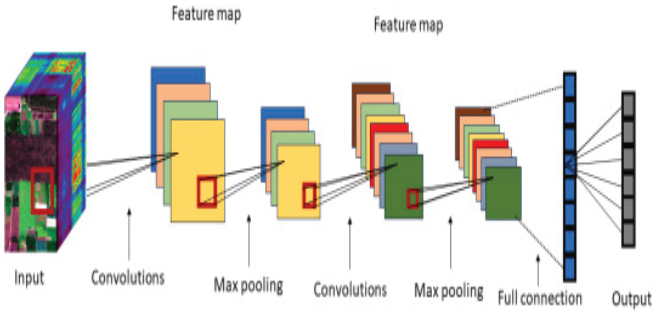

Fig. 2. Shows a graphical representation of a CNN architecture.

The yearly distribution of papers obtained from Scopus using the keywords "agriculture," "deep learning," and "hyperspectral imaging" for the years 2017–2023 is shown in Fig. 1. The graph shows a relatively small number of publications in 2017, which reflects the fact that DL techniques are still being applied to HSI in the agricultural industry. Since 2018, there has been a noticeable increase in research activity, indicating the growing interest in using DL to analyze HSI datasets in agriculture. Figure 1. The number of HSI articles published in the agriculture sector by year (source: Scopus) [30].

B. SAE(Stacked Auto Encoder)

A useful and adaptable tool in HSI, stacked auto-encoders (SAEs) provide a distinctive method for high-dimensional hyperspectral data feature extraction and representation learning. SAEs are essentially a kind of artificial neural networks that can compress and reconstruct hyperspectral data by learning to convert input data back to its original form after encoding it into a lower dimensional latent space. Significant advancements in the state of the art have resulted from their use in HSI. While maintaining crucial spectral and spatial information, SAEs in HSI can lower the dimensionality of data. SAEs provide data compression by automatically extracting pertinent features from hyperspectral cubes, which makes the data easier to handle for later analysis tasks

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including target recognition (Xie et al., 2013) and category (Lin et al., 2013) as well as target detection (Xie et al., 2019; Han et al., 2014). The final softmax layer or logistic regression are typically used to fine-tune the classification layer of the supervised model. After being trained separately, auto-encoders (AE) were incorporated into each SAE layer and played a crucial role in reconstructing the input data. To extract abstract features, the raw data needs to be recreated layer by layer. Since the features learned from one AE led the gradual training of the subsequent AE, the unsupervised pre-training phase made it easier to reduce reconstruction mistakes for each AE. The baseline parameters, which comprised weights and biases for every AE, were established as the first benchmarks for SEA after this pretraining phase. When labelled data was introduced as the supervisory cue, error backpropagation was used to refine the parameters of each layer, and the structure's parameters were updated using the stochastic gradient descent algorithm. Denoising auto-encoders (DAE) and stacked denoising auto-encoders (SDAE) were proposed by Vincent et al. (2008) and 2010 as additional enhancing options (Deng et al., 2023). Zhou et al. (2019a) outline a novel and reliable HSI classification framework that uses a CDSAE stands for Compact and Discriminative Stacked Autoencoder. There are two main phases to this framework. In the first stage, local Fisher discriminant regularisation was applied to the SAE's hidden layers in order to extract low-dimensional discriminative features.

C. DBN(Deep Belief Network)

Deep belief networks (DBNs) have emerged as a promising alternative in HSI due to its sophisticated spectral and spatial information representation and robust foundation for unsupervised feature learning. According to Hinton (2009), DBNs are essentially a kind of artificial neural network with several hidden layers. By extracting abstract and hierarchical features from hyperspectral data, they are able to extract valuable information from high-dimensional datasets. According to Ackley et al. (1985), each hidden layer in this structure is a Restricted Boltzmann Machine (RBM), and the characteristics that each RBM layer finds are used provides the initial input for the subsequent layer. In conclusion, the network's parameters are carefully adjusted using a supervised approach, incorporating a softmax classifier in the terminal layer. In hyperspectral data, DBNs can automatically identify and simulate complex spatial patterns and spectral fingerprints. By iteratively learning representations at different levels of abstraction, DBNs excel at feature extraction, which is crucial for a variety of tasks like classification (Li et al., 2014), target detection (Zhang et al., 2020b), and anomaly detection (Zhang et al., 2022a).

D. RNN(Recurrent neural Netork)

In HSI, RNNs have become a versatile and dynamic tool that provide a special method for investigating hyperspectral data cubes' spectral dimension. Hyperspectral time-series analysis is a good fit for RNNs, which are essentially a kind of artificial neural network with loops that enable it to interpret sequential data. Temporal dependencies in hyperspectral data can be captured by RNNs in HSI. RNNs are excellent in modeling and taking advantage of the dynamic characteristics of the scene under observation, as they consider the HSI data collecting sequence. For tasks like tracking the movement of objects or events across multiple spectral bands, this makes them very helpful and change detection, where it's critical to identify minute changes in a scene over time. They can learn and take advantage of their recurrent architecture reduces noise and enhances the quality of hyperspectral images by utilizing the spectral correlations in the data. In remote sensing, where air interference and sensor noise can degrade the quality of hyperspectral data gathered, this has significant applications. RNNs have becoming increasingly popular in HSI classification due to their ability to interpret hyperspectral pixels as sequential data (Mou et al., 2017b). However, significant issues such as gradients' explosion or disappearance frequently limit the effectiveness of conventional RNNs. To examine spectral sequences and their relationships, these characteristics are then processed through several LSTM layers. The SS-LSTM model is tested on three benchmark datasets once it has been thoroughly trained on a sizable HSI dataset. RNN use in HSI is not without its difficulties, though. The potential computational complexity is one of the main issues, particularly when working using hyperspectral data cubes with high resolution, which enables require a lot of memory and processing time. Additionally, because RNN efficacy is dependent on these decisions, choosing the right RNN architecture and tuning hyper parameters can be difficult. Because it might be difficult to comprehend the learnt temporal patterns and how they relate to certain spectral properties, there are still issues with RNN interpretability in hyperspectral data processing.

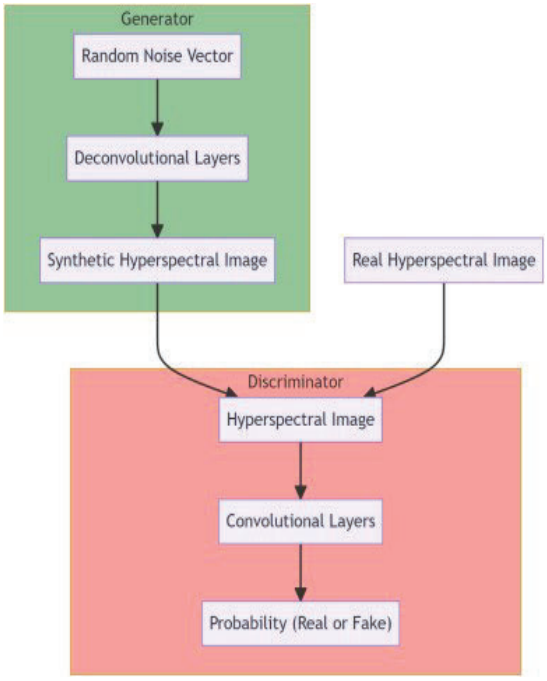

Fig. 3. A schematic representation of the GAN is shown in Fig. 3.

E. GAN(Generative Adversial Network)

Since the introduction of a dynamic interaction between two neural networks—the discriminator and the generator—GANs have become a novel and innovative method in HSI for producing, modifying, and improving hyperspectral data. Essentially, GANs may produce artificial hyperspectral images that closely mimic real-world data, creating new opportunities for super-resolution (Shi et al., 2022), denoising (Zhang et al., 2022c), and data augmentation (Zhang et al., 2022b). Hyperspectral data collection produces little-labeled

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information and is frequently expensive and resource-intensive. The HSI data undergoes a comprehensive analysis. For training, validation, and testing, the main HSI dataset is randomly divided into groups. The GAN network, which is intended to comprehend the underlying distribution and produce synthetic instances, then processes the training images. In the end, the larger training set's dimensionality is decreased using PCA. A Multi-Scale Residual Attention network (MSRA) receives the output data and uses it for training. To get the optimal training model, the validation set is used to update the network parameters on a regular basis.

F. AL(Active Learning)

The program selects the data points it uses in the active learning (AL) paradigm for learning on the fly. A standard that aims to select the most instructive samples serves as the basis for this selection procedure. In contrast to passive learning, which involves training the model on a dataset selected at random with no input during the selection process, the goal aims to enable the model to achieve higher accuracy while requiring less training examples (Brame, 2016). When there is a shortage of labelled training samples, this method helps to reduce costs and improve the quality of training samples related to acquiring a large number of labelled training samples. They presented an approach to classifying HSIs using enhanced AL in (Falahat Nejad and Karami, 2023). It is a two-step procedure that uses an augmented AL approach for both feature extraction and classification. Furthermore, to enhance the performance of classification of land cover classes, the authors of (Yadav et al., 2022) suggested Pixel-Certainty. Active Learning (PCAL) is a new AL approach. Two preprocessing techniques are Histogram Equalization (HE) and Distributed Intensity Filtering (DIF). Procedures used in the technique (Distante & Distante, 2020) to enhance image quality, after which textural information is extracted using an extended differential pattern (EDP) for feature extraction. With this knowledge, the PCAL approach classifies HSI patterns and shows resilience to changes in pixel noise.

V. FUTURE SCOPE & LIMITATIONS

Even though processing high band data presents many difficulties, researchers have advanced this field significantly over the past ten years, creating new methods and refining old ones. With the advent of machine learning and other technological developments, HSI classification has surpassed both state-of-the-art and conventional techniques in accuracy. Because of this, DL has become the most widely used tool for HSI classification in the last ten years, which has increased research into the properties of space imaging and remote sensing. New research avenues with a wide range of practical uses have been made possible by HSI exploration. Compared to multispectral and RGB imaging, HSI spectral bands have some advantages. For example, multispectral imaging only records images inside a small range of spectral bands, which may limit its capacity to identify particular chemical fingerprints or differentiate between different materials. HSI has drawbacks and restrictions as well. As a result, it is imperative to solve the issues, some of which are as follows:

- High Cost and Complexity of Data: One major obstacle is the complexity of processing high band data, which is further compounded by the huge financial ramifications. Making HSI more accessible can be aided by a thorough examination of the cost-benefit analysis and possible cost-cutting measures.
- Interpretation Difficulties Owing to Low Spatial Resolution: Although HSI data has extensive spectral information, it frequently has low spatial resolution, which causes irregularities and makes interpretation challenging. Future studies could concentrate on methods to improve spatial resolution or provide innovative approaches to interpretation that lessen this problem.
- Public Datasets with Limited Training Samples: The creation of reliable classification algorithms is hampered by the lack of large public training samples accessible datasets of HSI. Investigating methods for creating may be feasible to use cross-domain data or generated data.
- Target Detection Challenge: The creation of effective target detection algorithms is made more difficult by the unpredictability of the background and target spectra in HSI. To overcome this difficulty, future research could explore more sophisticated pattern recognition and machine learning methods.

The methods covered below provide researchers with fresh and enhanced methods for HSI analysis:

- HSI classification using meta-learning: The field of HSI classification has not fully tapped the potential of Meta learning, which entails developing algorithms with cross-dataset generalization capabilities. There is potential for creating more flexible and effective classification schemes in this field.
- Use of Newer Datasets: Using recently created datasets such as the olive, grape berry, and RIT-18 datasets can serve as a testing ground for both new and enhanced methods. Their application may result in categorization techniques that are more accurate and adaptable.

VI. CONCLUSION

This analysis concentrates on the developments in the use of HSI data in agriculture during the previous 20 years, with a particular focus on the incorporation using artificial intelligence (AI) methods to take advantage of this valuable data for automated and helpful purposes. In-depth evaluations of several approaches are provided, together with information on their effectiveness, research gaps, and accomplishments. It is notable for its thorough coverage of deep learning's evolution in this field. The work's overall uniqueness is increased by the important conclusions drawn from the analysis of each method. In addition to describing HSI technology uses in agriculture, the review offers a comprehensive appraisal of the state of HSI analysis research. It also presents recently created methods that might have an impact on further studies in the area. An unparalleled glimpse into the complexities of farming variability is made possible by the breadth and depth of data that HSI provides. HSI offers a wide range of information, from identifying possible insect infestations to comprehending even the smallest changes in soil composition. However, sophisticated processing methods are required because of the vast amount and complexity of this data. AI—more especially, DL—comes into play here. AI is the ideal complement to HSI's data-rich outputs because of its

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capacity to sort through enormous databases, spot trends, and produce useful insights. When combined, they provide a powerful blend that can tackle some of the most important issues facing agriculture. In addition to the technical elements, a larger socioeconomic factor should also be taken into account. It is crucial to make sure that the advantages of this technological convergence are shared fairly as AI and HSI are increasingly incorporated into agriculture.

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